Land-Use Land-Cover Prediction from Satellite Images using Machine Learning Techniques

Tapan Kumar Das[1,#], Senior Member, IEEE, Dillip Kumar Barik[2], K V G Raj Kumar[3]

[1]*School of Information Technology and Engineering,Vellore Institute of Technology, Vellore, India*
[1]tapan.das@vit.ac.in
[2]*School of Civil Engineering,Vellore Institute of Technology,Vellore, India*
[2]b.dillip@gmail.com
[3]*DE & S Campus,Telangana State Remote Sensing Application Centre, Hyderabad, India*
[3]drrajkumar0919@gmail.com

Abstract— **The objective of the proposed research is to estimate the land-use/ land-cover (LULC) changes by employing artificial intelligence techniques rather than doing it manually. For this purpose, Sentinel-2 satellite images are used; these images are overtly and can be accessible readily within the Earth observation Copernicus. Sentinel-2 images labelled as EuroSAT data are employed, these images cover 13 spectral bands and consist of ten categories. The presented model will ease the process of classification of images so that various usage types of lands will be revealed out of this. In this paper, classification using supervised machine learning (ML) techniques e.g. Random Forest, K- Nearest Neighbor (KNN), Support Vector Machine, Decision Tree, Gradient booster and Ensemble classifiers by stacking all these models are carried out. Furthermore, the results of all the six models supported by the metrics like accuracy, precision, F1 score, and recall are compared. Finally, it is identified that Ensemble classifier is the highly efficacy model which may be applied for classifying LULC cover in order to achieve a highly accurate result in ground data.**

Keywords—land use/land cover (LULC); classification; machine learning

I. INTRODUCTION

The satellite geographical data are immensely useful for the prediction and analysis of numerous applications e.g. weather forecast, cropping and yield, disaster prediction , climate changes, urbanisation and so on. One such fundamental application is LULC. In this field, one major task is automatically characterizing the land basing on the physical land type and related usage. Generally, the land cover signifies how the type of land is being developed recently e.g. forest, water, concrete, on the other hand land use documents how people are using the land and Land use shows how people use the landscape that is, whether for development, conservation, or mixed uses. LULC classification is an investigation where we can easily differentiate land cover variables and can perform various actions accordingly. It is mostly desired for future prediction of LULC type for sustainable habitation and development.

With the advent of artificial intelligence such as machine learning, it has been possible to classify a given data into possible categories by using specific algorithms. It is pertinent from many observations that the performance and efficiency of ML models depends upon the trainable labelled data and the quality of the classes we are using. Hence, EuroSAT dataset is selected; EuroSAT is an image dataset spread over ten unique classes consisting of 64x64 sized 2700 hyper spectral images covering 13- spectral bands. The EUROSAT dataset is composed of images retrieved from the Sentinel-2 satellite.

Hyper spectral images are analysed based on spectroscopy. Spectral resolution arises in hyper- spectral images only. In this paper, we demonstrate different machine learning model's performance with respect to land usage and land covering classification and hence based on performance and efficiency, the most apposite model is identified. For the simulation of the classification algorithms, we have used publicly available EuroSAT dataset.

II. RELATED STUDY

Usage of deep learning techniques i.e. Deep CNN, ResNet and Google Net in processing satellite images for LULC classification has been studied [1] and they have used metrics such as accuracy and precision for the result evaluation. Another study employed Markov random fields, expectation-maximization (EM) algorithm, support vector machines, spatial correlation for predicting the land classes by ground- truth information [2]. Moreover, they have used both supervised and unsupervised learning mechanisms; for analysis, they have used metrics such as accuracy, correlation matrix and average gap.

Phartiyal et al [3] classified the land cover in mixed land cover scenarios with PolSAR data. Polarimetric decomposition techniques are most popular methods for PolSAR data classification in recent times. These techniques focus on identification of dominant scattering phenomena and hence result in sub-optimal classification in mixed land cover scenarios. Alternatively, polarization signatures (PSs) are good illustrations of SAR target responses as they depict a detailed physical information from target backscatter. Researchers have successfully utilized SAR PSs for land cover (LC) classification. Some reports suggested utilizing correlation between observed PSs and standard target PSs as features for LC classification. This paper presents a study on improved utilization of PSs for optimal LC classification in mixed

978-1-6654-9602-5/22/$31.00 ©2022 IEEE

class scenarios. First, PS based SAR features are derived using fully polarimetric SAR data. The features represent a degree of similarity between observed and standard PSs. The derived features are termed as polarization signatures correlation features or PSCFs. The novel PSCFs are analyzed, evaluated and compared with decomposition based features for the purpose of LC classification.

The potential of repeat-pass interferometric SAR (InSAR) for classifying land cover has also been explored [4].

Ghorbanian et al. [5] in their study utilized multiple sources of data along with object-based Random forest (RF) techniques and contrasted the performances of Sentinel-1 with Sentinel-2 datasets. However, they used object-based classification rather than pixel- based and used metrics as Overall Accuracy (OA), Kappa coefficient. Efficiency, time and cost for analysis.

Ge et al [6] used machine learning standard classifiers for LULC classification and achieved an overall accuracy of 90% and concluded that on further processing, the accuracies could be enhanced.

Emparanza et al [7] used CNN methodology and multi-class models for land-cover classification. The uniqueness of this study is authors trained and tested using two different classes, and achieved more accurate prediction.

Dong et al [8] utilized optimal geometric feature with adaptive neighborhood, k-means clustering, multi-dimensional scaling for classification of urban land. They enforced a novel feature selection method based on correlation coefficient, multidimensional scaling and k-means clustering produced better results. Xu et al [9] used multi-temporal image couple with SVM and 3D- CNN for land cover classification. In this paper, a dynamic architecture based classification is being enforced. the hierarchy of additional complicated options are extracted and supported CNN based stacking model over the whole feature area.

III. MACHNE LEARNING ALGORITHMS

A. K-Nearest Neighbor (KNN)[10]

Initially K value is determined by user. When an input is provided, it fetches the Euclidean distance from training data and basing on the minimum distance, the kth neighbor is identified. Furthermore, the class label for those neighbors are assigned on majority voting. For land cover data length, tallness, bandwidth from the images serve as the input.

B. Support Vector Machine

The novelty of Support Vector Machines (SVMs) [11] is it can process a high dimensional space of features and identify underlying contrasts between classes of given information.

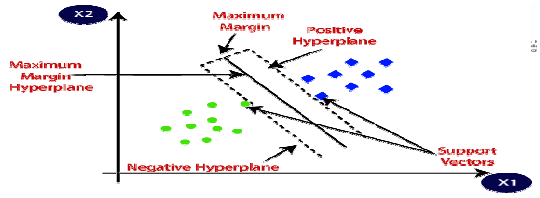

Fig. 1 SVM Classifier

Moreover, linear SVM partition the input data to either of the classes and a typical model is shown in Fig.1.

C. Random Forest(RF)[12]

It is a type of supervised learning techniques which symbolizes ensemble learning paradigm by packaging several decision trees and relying its output on underlying output of decision trees' and moreover the output is expected to be better than individual trees. Random decision forests right for choice trees' propensity for over fitting to their preparation set. Fig. 2 epitomize the similarity between random forest and decision tree.

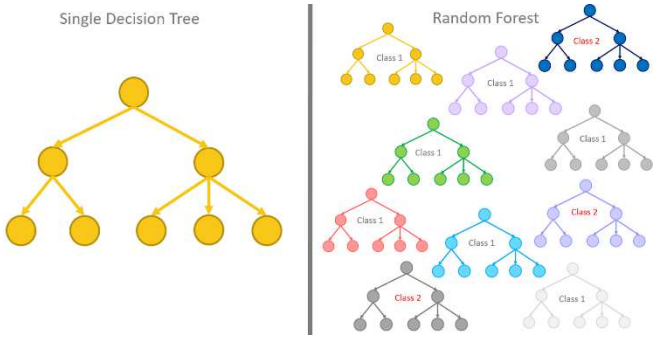

Fig. 2 Decision tree vs Random forest

D. Decision Tree (DT)

A Decision tree is a conceptualize the binary tree design in which each interior hub addresses a test on a trait, each branch signifies the result of the preceding test, and each leaf hub represents addresses a class name. The walk from root to leaf characterizes a rule.

E. Gradient Boosting(GB)

This technique creates a forecast model as a gathering of feeble expectation models, normally choice trees. When a choice tree is the frail student, the subsequent calculation is called inclination helped trees, so that it beats arbitrary woodland. It constructs the model in a phase astute style and sums up by permitting enhancement of a self-assertive differentiable misfortune work.

F. Ensemble Classifier[13]

Ensemble classifier helps in improving machine learning results by combining several models. As compared to a single classifier, a better performance can be achieved by combining multiple classifiers. In this context, a set of classifiers are made to learn and next they are stacked in order to have a sequential upgraded learning and this is represented in Fig. 3. Here in this work we use stacking Ensemble classifier which basically stacks all the five algorithms used; KNN, Random forest, Decision Tree, SVM, Gradient booster one on another in which each algorithm helps boosting the other and produces better results.

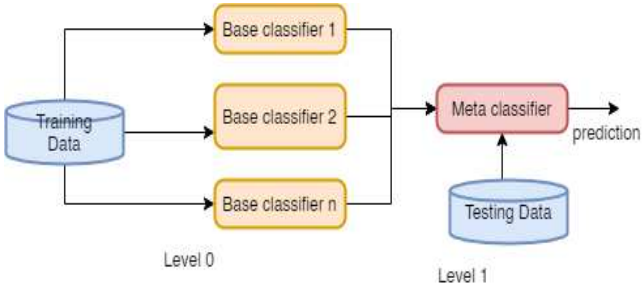

Fig. 3 Stacking based Ensemble classifier

IV. METHODOLOGY

In this section, EuroSAT data is utilized[14] by employing the following algorithms i) Random Forest ii) K-nearest neighbor, iii) Support vector machine, iv) Decision tree, and v) Gradient booster. Further ensemble voting classifier is proposed in order to achieve a better land cover classification results as compared to their peers. The detailed methodology is depicted in Fig.4.

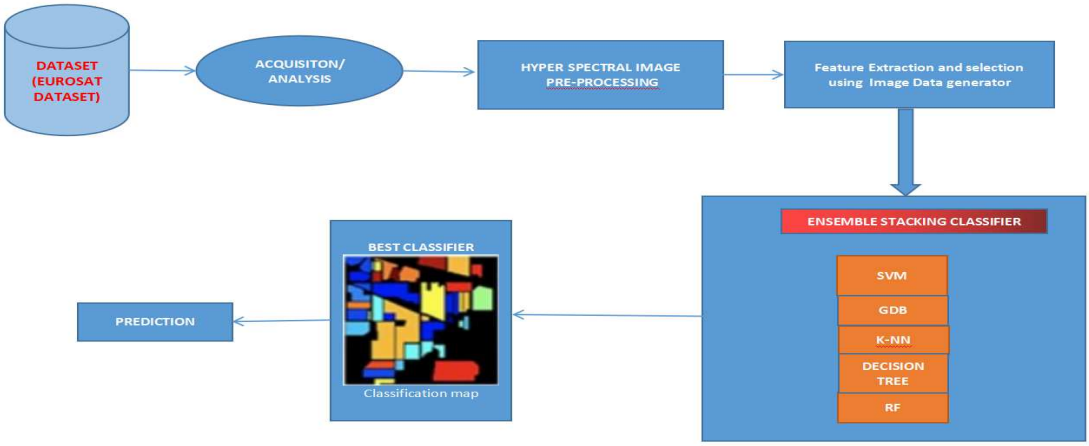

Fig. 4 Detailed Flow of Methodogy

A. Data Exploration

For this research, EuroSAT dataset is analysed which consists of images captured by the Sentinel-2 satellite. The constituting images are being labelled into 10 different categories. The dataset consists of 64x64 images captured by Sentinel-2A satellite; it has over 27000 images spread across 10 classes, and the detailed classes are represented in Fig.5. The data consists of hyperspectral images with 13 spectral bands.

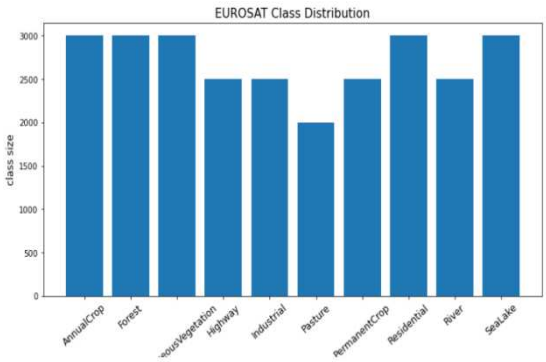

Fig. 5 Class distribution of EuroSAT data

From Fig 5, it is obvious that LULC classes are annual crop, permanent crop, residential building, industrial building, river, sea and lake, highway, pasture, vegetation and forest.

B. Hyper Spectral Image Preprocessing

Prior to analysis of hyper spectral images, images are being pre- processed for removing undesired spectral artifacts that might affect the classification result. This step includes noise removal, light scattering correction and highlighting of spectral derivatives.

C. Classification using Machine Learning Models

First, it is checked as how a machine learning model performs by directly feeding it the whole image pixels (64x64x3 = 12288). Out of 27000 samples dataset the training data is of size 21600 and test data is of size 5400. Authors used a trick to utilize Keras Image Data, the generator feature transform the image dataset to a numpy array and presented this classification by demonstrating machine learning models.

D. Comparative Analysis

Finally, a comparison among all the six models is exhibited to find out the model generating the more accurate results. Thereafter by employing different models, including ensemble classifier, we can generate contrasting land cover images.

V. EXPERIMENTAL RESULTS

In this section, experiments are executed for LULC classification using RF, KNN, SVM, DT, GB and ensemble of these classifiers. In our application, 27000 hyper spectral images captured using sentinel-2 satellite pictures with 13 spectral bands and 10 different classes. These input data were divided in 80:20 ratio for testing and training with 21600 and 5400 images respectively. These training and testing datasets were processed through machine learning algorithms using ensemble classifier and brought about a maximum accuracy of 98.91% accuracy, precision of 0.99 . However, SVM also yielded

comparative result of accuracy of 96.58%. Subsequently GB also exhibited an accuracy of 94%.

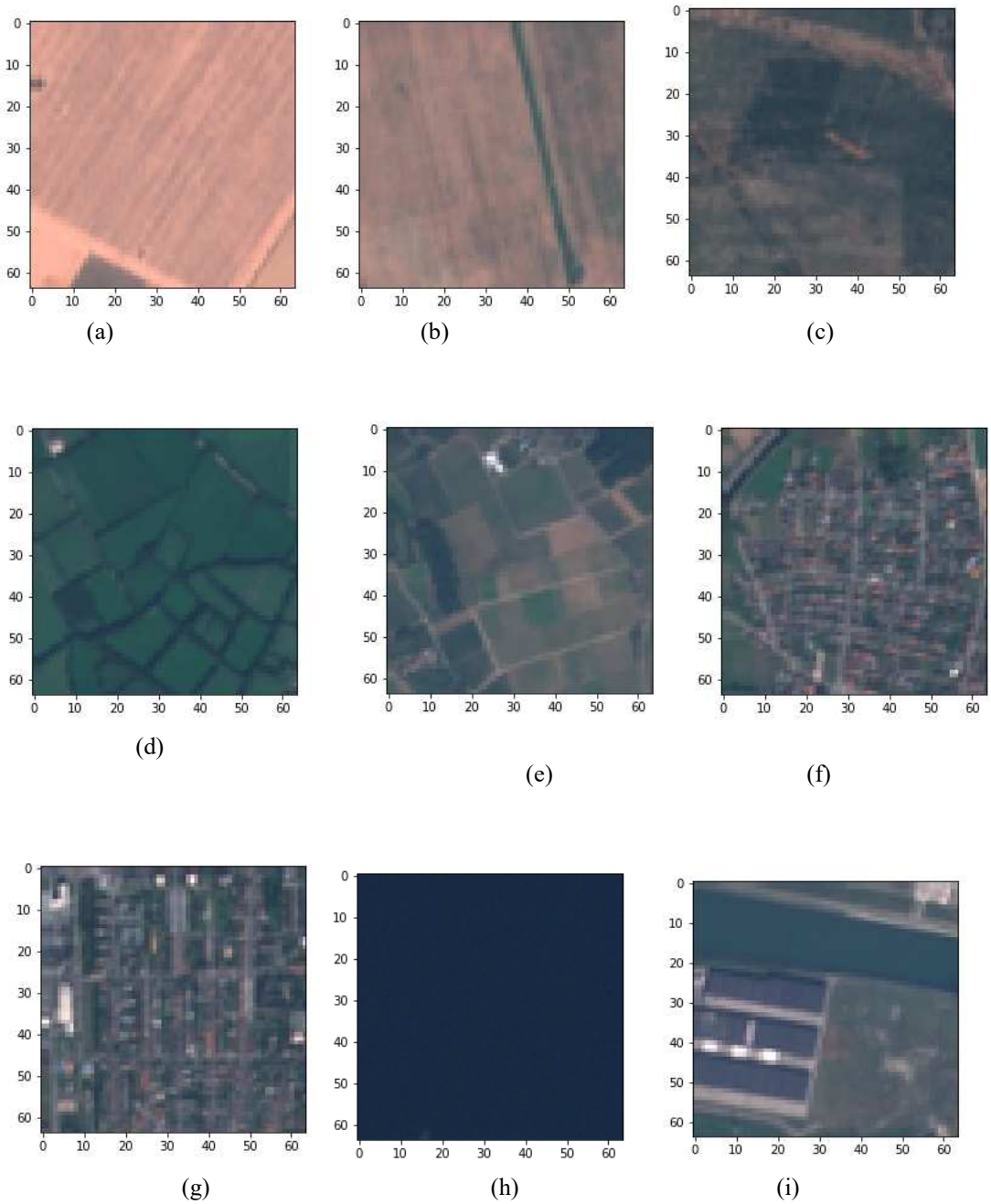

Fig. 6 Predicted output with Ensemble classifier (a) Annual crop (index 0) (b) Annual crop (index 0) (c) Herbaceous vegetation (index 2) (d) Pasture (index 5) (e) Permanent crop (index 6) (f) Residential (index 7) (g) Residential (index 7) (h) Sea/ Lake (index 9) (i) River (index 8)

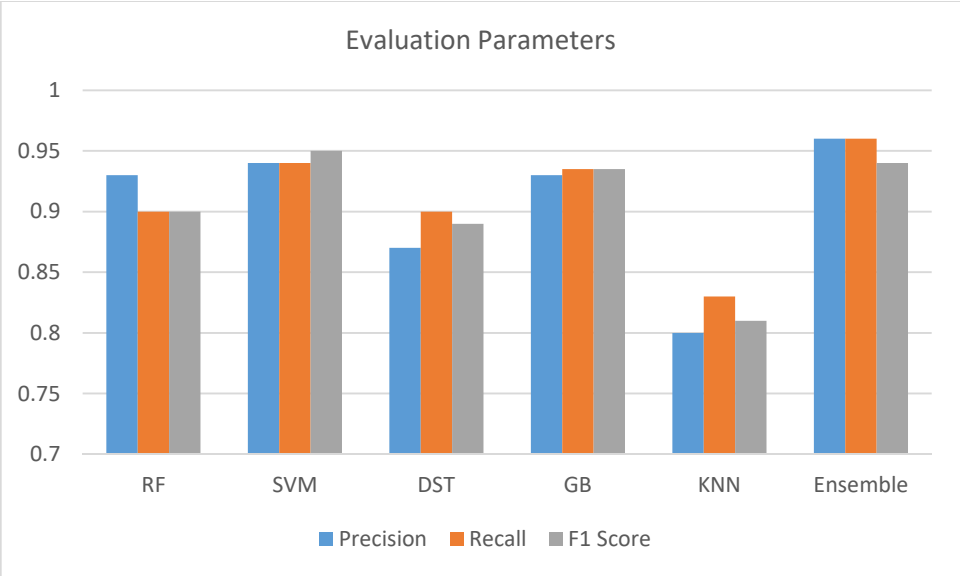

Fig. 7 Precision, Recall and F1 score of all the six classification techniques

:Fig. 7 represents precision, recall and F1-score attained by all the six classifiers used in this study while Fig. 8 exhibits their individual accuracies in classifying the land cover type.

TABLE 1 QUALITY PARAMETERS

LULC Type	P	R	F1
Annual crop	0.958	0.951	0.955
Forest	0.990	0.991	0.990
Herbaceous vegetation	0.985	0.956	0.970
Highway	0.985	0.983	0.984
Industrial building	0.986	0.945	0.965
Pasture	0.969	0.988	0.978
Permanent crop	0.935	0.968	0.951
Residential crop	0.966	0.998	0.982
River	0.985	0.985	0.985
Sea/Lake	0.998	0.989	0.993

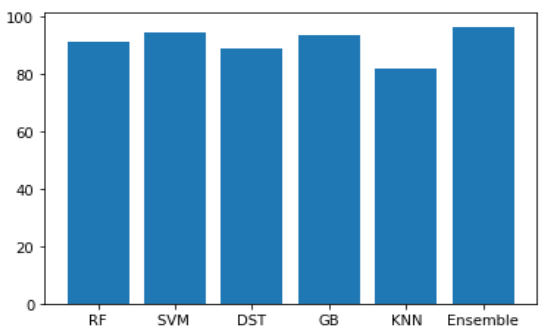

Fig. 8 Accuracy of classifiers (Histogram)

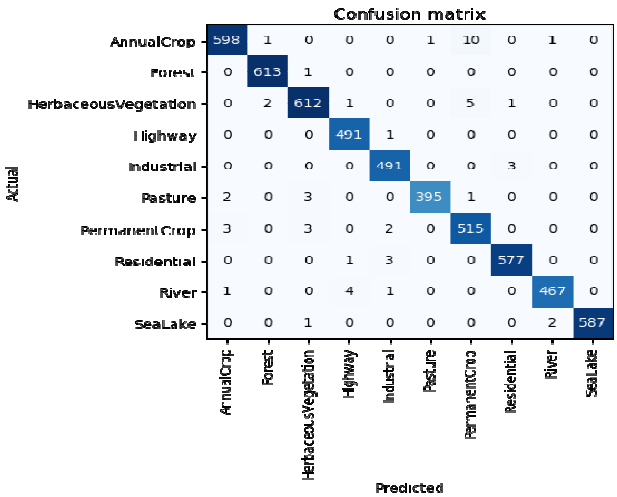

Fig. 9 Confusion Matrix

The confusion matrix in Fig.9 displays the contrast between actual and predicted output. Hence, four classes emerge; they are i) True positive (T_{pos}), ii) False positive (F_{pos}), iii) True negative (T_{neg}) and iv) False negative (F_{neg}). Using these values , quality parameters , Precision (P), Recall (R) and F1-score (F1) [15] for the different types of land cover type is exhibited in Table 1.

VI. CONCLUSION

In this paper, an attempt has been made to automatically trace the land use pattern from remote sensing images as it is quite hard to do it manually by surveying the ground data. In doing so, several machine learning models are implemented; next an ensemble classifier encompassing KNN, DT, RF, SVM and GB classifiers is built and the results produced by each algorithms are reported. On the basis of these results, it is concluded that SVM produces best classification results and the hybrid ensemble yields better than individual algorithms. In future, a higher accuracy in classification can be attained by using hybrid

machine learning algorithms and optimizing the input parameters.

REFERENCES

[1] P. Helber, B. Bischke, A. Dengel, and D. Borth, "Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol.12, no.7, pp. 2217-2226, 2019.

[2] M. Marconcini and D. Fernàndez-Prieto, "A novel approach to targeted land-cover classification of remote-sensing images." In 2012 IEEE International Geoscience and Remote Sensing Symposium, pp. 7345-7348. IEEE, 2012.

[3] G. S. Phartiyal, K. Kumar, and D. Singh, "An improved land cover classification using polarization signatures for PALSAR 2 data," Advances in Space Research, vol. 65, no.11, pp. 2622-2635.2020

[4] F. Sica, A. Pulella, M. Nannini, M. Pinheiro, and P. Rizzoli, "Repeat-pass SAR interferometry for land cover classification: A methodology using Sentinel-1 Short-Time-Series," *Remote Sensing of Environment*, vol. *232*, 111277, 2019.

[5] A. Ghorbanian, M. Kakooei, M. Amani, S. Mahdavi, A. Mohammad Zadeh, and M. Hasanlou, "Improved land cover map of Iran using Sentinel imagery within Google Earth Engine and a novel automatic workflow for land cover classification using migrated training samples," ISPRS Journal of Photogrammetry and Remote Sensing, vol.167, pp.276-288, 2020.

[6] G. Ge, Z. Shi, Y. Zhu, X.Yang, and Y. Hao, "Land use/cover classification in an arid desert-oasis mosaic landscape of China using remote sensed imagery: Performance assessment of four machine learning algorithms," Global Ecology and Conservation, vol.22, p.e00971, 2020.

[7] P. R. Emparanza, N. Hongkarnjanakul, D. Rouquette, C. Schwob, and L. Mezeix, "Land cover classification in Thailand's Eastern Economic Corridor (EEC) using convolutional neural network on satellite images," Remote Sensing Applications: Society and Environment, vol. 20, p.100394,2020.

[8] W. Dong, J. Lan, S.Liang, W. Yao, and Z. Zhan, "Selection of LiDAR geometric features with adaptive neighborhood size for urban land cover classification," International journal of applied earth observation and geo-information, vol.60, pp.99-110, 2017.

[9] Z. Xu, K. Guan, N. Casler, B. Peng, and S. Wang, "A 3D convolutional neural network method for land cover classification using LiDAR and multi-temporal Landsat imagery," ISPRS journal of photogrammetry and remote sensing, vol.144, pp.423-434,2018.

[10] T. K. Das , "A Customer Classification Prediction Model Based on Machine Learning Techniques," 2015 International Conference on Applied and Theoretical Computing and Communication Technology (iCATccT), 29-31 Oct , 2015. pp. 321-326.Davangere, India, 2015.

[11] S. Baamonde, M. Cabana, N. Sillero, M.G. Penedo, H. Naveira, and J.Novo, "Fully automatic multi-temporal land cover classification using Sentinel-2 image data," Procedia Computer Science, vol. 159, pp.650-657, 2019.

[12] P.K. Roy, A. Singh, A.K. Tripathy, and T. K. Das," Identifying Cyberbullying Post on Social Networking Platform Using Machine Learning Technique." In Advances in Distributed Computing and Machine Learning (pp. 186-195). Springer, Singapore, 2022.

[13] T. Aboneh, A. Rorissa, and R. Srinivasagan, "Stacking-Based Ensemble Learning Method for Multi-Spectral Image Classification." Technologies, 10(1), p.17, 2022

[14] P. Helber, B. Bischke, A. Dengel,. and D. Borth, "Introducing eurosat: A novel dataset and deep learning benchmark for land use and land cover classification," In IGARSS 2018 IEEE International Geoscience and Remote Sensing Symposium (pp. 204-207). IEEE, 2018.

[15] P. K. Roy, A. K. Tripathy, T. K. Das and X. Z. Gao, "A Framework for Hate Speech Detection Using Deep Convolutional Neural Network," IEEE Access, Vol. 8, pp. 204951-62, 2020.

.